Federated Learning-Based Portfolio Rebalancing Necessity Detection

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Abstract*—Portfolio rebalancing is usually governed by fixed calendar or drift-band rules that ignore the market state in which the decision is taken, and the machine learning systems proposed to replace those rules are commonly trained on labels emitted by the very heuristics they are meant to supersede. Because such a label is a deterministic function of the input features, the resulting classifier recovers the rule rather than the decision, an effect this paper names rule-oracle degeneracy and measures directly. We present a federated learning-based system that treats rebalancing as binary necessity detection over a versioned twelve feature portfolio and market contract, labelled by the sign of a 21-trading-day, net-of-cost certainty-equivalent gain evaluated over 3,706 NIFTY 50 sessions spanning 2011 to 2026. Detection and federation are developed as equal parts of one design: a logistic detector whose parameters are a twelve-element weight vector, and a sample-weighted federated averaging protocol with L_2-clipped updates, contract gating and atomic activation, in which model updates rather than holdings leave the client. The detector attains a test ROC-AUC of 0.6489 (five-fold cross-validation 0.6519 ± 0.0094) with an expected calibration error of 0.0214. Relabelling the identical 30,000 states with the project's own heuristic, changing nothing but the label function, raises test ROC-AUC to 0.9890, a gap of 0.3401 attributable to the label and not to the learner, across which the correlation between elapsed time and the label rises from 0.0256 to 0.6074. That heuristic predicts realised net-of-cost benefit at 0.5278, and calendar and band rules at 0.5167 and 0.5146, all close to chance. The aggregation path is implemented and unit-tested but disabled by default and not experimentally evaluated here, and no privacy guarantee is claimed.

***Index Terms*—portfolio rebalancing, necessity detection, federated learning, federated averaging, logistic regression, model calibration, NIFTY 50**

I. Introduction

A portfolio left untouched does not stay where it was put: differential asset returns drift realised weights away from the risk–return position that was chosen [1], [2]. Rebalancing restores that position, but every restoration pays spread, brokerage and impact costs and liquidates positions whose recent strength may persist [3], so the decision is a cost–benefit test on the current drift, market state and transaction cost jointly. Incumbent practice answers with rules that use almost none of this information: calendar rebalancing acts on a fixed interval and drift-band rebalancing on a fixed threshold, both blind to the volatility, drawdown and trend that determine whether a round trip is worth its price. The applied literature argues this conceptually rather than measuring it [4], [5]; measured against forward, net-of-cost outcomes on our data, a 90-day calendar rule and a drift-band rule attain test AUC 0.5167 and 0.5146, both close to chance. Fig. 1 shows the system in which we replace such a rule with a learned detector.

The data this decision needs cannot reasonably be centralised: holdings disclose wealth, leverage and trading history and remain identifying long after any label is stripped, while a single participant produces too few rebalancing states to learn a market-wide regularity alone. Federated averaging resolves this cleanly for a linear detector, since its parameters — a twelve-element weight vector and intercept — are exactly the object federated averaging is defined over [6], needing no distillation step. Transmitting updates instead of holdings is data minimisation, not a privacy guarantee; no differential privacy or secure aggregation is implemented anywhere in this paper.

Supervised rebalancing has no naturally observable label, and the expedient response is to label historical states with the output of the very heuristic the model should replace. We call the resulting pathology *rule-oracle degeneracy*: the label is a deterministic function of the features the rule reads, so the classifier decompiles the rule instead of learning an economic decision. Section IV shows this is large enough to be mistaken for success: on one fixed corpus, holding everything but the label function constant, the same pipeline reaches test AUC 0.9890 against heuristic labels and 0.6489 against a forward, net-of-cost backtest.

A. Contributions

- **C1: Backtest-derived labels for rebalancing-necessity detection.** Each label is the sign of a forward 21-trading-day, net-of-cost certainty-equivalent gain (Section III), evaluated on 3,706 NIFTY 50 sessions (2011-08-01–2026-07-31) at 0.35% round-trip cost, over 30,000 generated states with structural positive rate 0.5009.
- **C2: A quantified measurement of rule-oracle degeneracy.** A matched control varying only the label function measures a gap of 0.3401 test AUC attributable to the label, not the learner (Section IV).
- **C3: A hardened federated aggregation protocol.** Updates are L_2-clipped and combined by a sample-weighted mean, gated by a minimum participant count, an upload cap and a per-account throttle, with atomic activation. It is implemented and unit-tested but disabled by default and not experimentally evaluated here.
- **C4: A cross-runtime parity contract** that pins feature order and scaler constants by version, so pooling is well defined and mismatched contracts are rejected rather than averaged, to $10^{-12}/10^{-5}$ agreement.
- **C5: An explicit model-selection trade.** The linear detector surrenders 0.0301 test AUC to a random forest and is chosen anyway because its parameters are averageable and its probabilities are calibrated (ECE 0.0214); the deficit is reported, not suppressed.

The evidence is correspondingly narrow: the cross-section is generated rather than observed, no path-based backtest or federated experiment was run, and none of this constitutes investment advice. Section II surveys the related literature; Section III presents detection and federation as equally weighted halves of one design; Section IV reports the results; Section V concludes.

II. Literature Survey

Rebalancing policy. Portfolio construction selects weights that trade expected return against variance [1], and risk-adjusted performance measures make the resulting drift consequential [2]. Practice answers drift with calendar or threshold rules, neither free of opportunity cost since restoring target weights trims positions whose intermediate-horizon returns are reported to continue rather than revert [3]. The applied literature that proposes machine learning here argues the point conceptually: Adedoyin catalogues the model families that might replace fixed-interval rebalancing and the overfitting risk they face [4], and Kumari reaches a similar conclusion for threshold policies, naming logistic regression as the appropriate device for a binary financial outcome [5]. Both are reviews; neither states a reproducible method, a labelling procedure or a numeric result, and this work uses them only as motivation and as precedent for the model class.

Financial ML and federated mechanics. Mashrur *et al.* survey machine learning for financial risk management and establish supervised classifiers, logistic regression among them, as standard instruments for binary financial determinations, while also naming generated data and federated learning as directions for settings where confidential or rare-event data are unavailable [7]. Deep architectures apply the same representation-learning spirit to asset universes [8]. Neither models the portfolio-level rebalancing action itself. Federated averaging trains a shared model by repeatedly averaging local updates instead of centralising data [6], applicable across the horizontal, vertical and transfer settings surveyed by Yang *et al.* [9]; its dominant obstacle is statistical heterogeneity, which degrades convergence under non-independent client data [10], [11], and surveys of federated recommendation add that retaining data on-device removes a class of collection risk while introducing inference and poisoning threats of its own [12].

Financial federated evidence. Noseda *et al.* train a shared volatility classifier and report federated averaging reaching accuracy close to centralised training under an IID partition (0.7162 vs. 0.7295) and exceeding isolated local training under a non-IID partition (0.7206 vs. 0.5366), while strong differential-privacy noise collapses utility below every non-private configuration [13]. Yu, Wynter and Lim apply federated reinforcement learning to portfolio management and find that plain aggregation causes training to fail, motivating a personalised, regularised server objective [14]. The lesson is that collaboration can approach centralised quality on heterogeneous financial data, but neither privacy mechanisms nor plain averaging are free of cost.

Table I positions the present work against this survey. No identified work treats whether a drifted portfolio requires rebalancing now as supervised binary classification trained on forward, net-of-cost outcome labels, and none measures the degeneracy that arises when a rebalancing model is trained on the labels of the rule it replaces. This paper occupies that slot: it formulates rebalancing-necessity detection on backtested outcome labels, quantifies the resulting degeneracy against heuristic labels, and couples the detector to a federated design exchanging model updates rather than portfolio records.

III. Methodology

This section presents the system in two equally weighted halves: detection and federation.

A. System and Problem Formulation

The detector is embedded in the three-tier system of Fig. 1: a *browser tier* that builds the feature vector and infers locally, a *platform tier* that holds identity and the aggregation core and never receives holdings, and a *model tier* that is the single authoritative trainer. Two versioned contracts hold the tiers together — a feature/scaler specification and a set of parity fixtures — detailed in Section III-E, since they are preconditions for aggregation rather than incidental engineering. Only a twelve-element coefficient vector, a scalar intercept and a sample count ever cross a tier boundary.

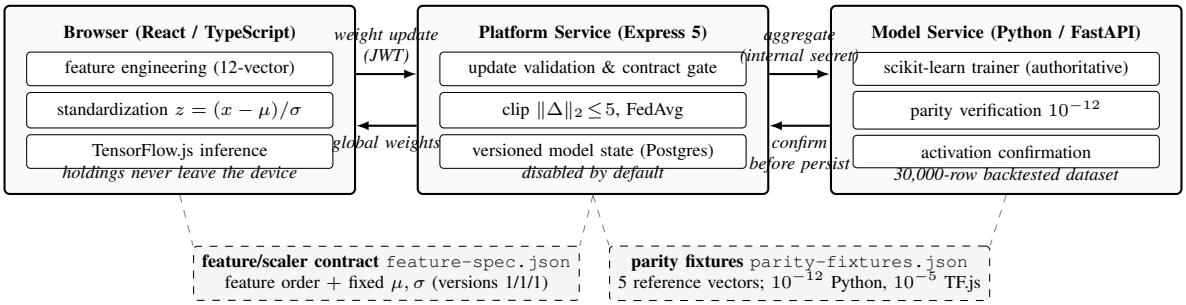


Fig. 1. System architecture. Features are computed and standardized in the browser; the platform validates and aggregates federated updates; the model service is the authoritative trainer that confirms an aggregate before it is published. Two versioned contracts keep both runtimes numerically identical. The federated path is implemented and unit-tested but disabled by default and not evaluated in this paper.

TABLE I
POSITIONING OF THE PRESENT WORK AGAINST THE SURVEYED LITERATURE

Work	Primary focus	Method	Federated	Label source	Numeric results
Adedoyin [4]	Rebalancing	Conceptual, model-agnostic	No	—	None reported
Kumari [5]	Portfolio management	Conceptual review	No	—	None reported
Mashrur *et al.* [7]	Financial risk	Survey (taxonomy)	Named as future work	—	Survey only
Heaton *et al.* [8]	Portfolio construction	Deep architectures	No	Market returns	Yes
Noseda *et al.* [13]	Volatility forecasting	LSTM with FedAvg	Yes, evaluated	Engineered volatility	Yes
Javeed *et al.* [12]	Recommender privacy	Survey (threat model)	Yes, as subject	—	Survey only
Yu *et al.* [14]	Portfolio management	Federated reinforcement learning	Yes, evaluated	Reinforcement reward	Yes
This work	**Rebalancing necessity**	Logistic detector with FedAvg	Implemented, not evaluated	**Forward net-of-cost outcome**	Yes

The two columns that isolate the gap are the last two. No surveyed work poses rebalancing necessity as a supervised binary decision, and none derives its labels from a forward, net-of-cost outcome; where labels exist they are engineered targets or reinforcement rewards. The federated column is reported honestly: the aggregation path of this work is implemented and unit-tested but is not experimentally evaluated here, whereas [13] and [14] report federated experiments.

Let $x \in \mathbb{R}^{12}$ describe a portfolio and its market context on a decision day. Rebalancing-necessity detection is the binary task

$$\hat{y} = g\big(f(x;\theta)\big), \qquad y \in \{0,1\}, \tag{1}$$

where $y = 1$ denotes that rebalancing is warranted. This models the *decision*, not the trade schedule implementing it, which is a separate deterministic engine outside this scope.

B. Features, Dataset and Labels

The contract fixes twelve quantities in order: seven portfolio descriptors, one elapsed-time descriptor and four market descriptors, defined together with their distribution and univariate association with the label in Table II. The vector is assembled in that fixed order and standardized by shared constants,

$$z_i = (x_i - \mu_i)/\sigma_i \;\; \text{if } \sigma_i \geq 10^{-9}, \;\; \text{else } 0, \tag{2}$$

part of the versioned contract rather than fitted per run, since federated averaging is coherent only when every participant occupies the same input space. This carries a disclosed defect: the shipped constants derive from a legacy 5,000-row dataset rather than the 30,000-row training split used here (largest discrepancy on `num_stocks`, $|\Delta\mu|$=3.9504); the defect is disclosed rather than repaired, and Section IV quantifies its cost at $+0.0001$ test AUC.

No public dataset pairs portfolio states with realized rebalancing outcomes, so the corpus is constructed — generated data is established practice in financial machine learning where labelled outcomes are unobtainable [7]. The generator emits 30,000 states under seed 20260804; each row is one simulated decision on a real trading day. The *market leg is real*: NIFTY 50 daily closes over 3,706 sessions (2011-08-01 to 2026-07-31) supply every return and market feature, while only the cross-section of holdings is generated from one of six basket profiles with log-normal target weights and a persistent per-symbol factor loading $\beta_i \in [0.55, 1.55]$. The book drifts over 5–170 trading days by a real market return plus a systematic, sector and idiosyncratic shock, and forward returns combine the real forward index move with a regime-dependent reversion term, stronger under volatility/drawdown stress and weaker in calm uptrends [3]. Persistent loadings keep the market factor from averaging out, regime-dependence keeps returns from collapsing into a function of weights alone, and drift is *caused* by elapsed time, so elapsed time reaches the label only along that causal path; exact constants are in the repository alongside the reproduction script.

The label is the outcome of a forward, net-of-cost backtest rather than a rule. Let t be target weights, w the drifted current weights and f the realized forward returns. The investor compares rebalancing against holding under a mean-variance objective with transaction costs [1], [2], and the label is the sign of the resulting net certainty-equivalent gain,

$$\begin{aligned}\Delta\mathrm{CE}(w,t,f) = (t-w)^{\top} f + A\big[\mathrm{Var}(w) - \mathrm{Var}(t)\big] \\ - c\,\|t-w\|_1, \qquad y = \mathbf{1}\{\Delta\mathrm{CE} > 0\},\end{aligned} \tag{3}$$

with risk aversion A=45.0 and round-trip cost c=0.0035 (35 basis points): a *return gain*, a *risk gain* priced through ex-ante

variance, and an *implementation cost*. Because cost scales with turnover, (3) reproduces no-trade-band behaviour without any band being coded. Over 30,000 rows it yields a positive rate of 0.5009 obtained structurally, without resampling. The elapsed-time feature τ is deliberately absent from (3); it influences the label only through the drift it causes, removing the time-collinearity defect of the predecessor heuristic, which instead scored a sigmoid combination of a logistic function of τ and a tiered adjustment on drift, concentration, drawdown, trend and volatility directly from the feature vector. Because that score is a deterministic function of the same features the classifier receives, a model trained on its output recovers the rule rather than the decision; it is used here solely as a negative control, never as the reported detector's label source.

C. Detection Model and Training

The detector is a binary logistic regression on the standardized vector of (2),

$$P(y = 1 \mid x) = \sigma\big(\langle \mathbf{w}, z \rangle + b\big), \quad \hat{y} = \mathbf{1}\{P \geq 0.5\}, \qquad (4)$$

fitted by L_2-regularized maximum likelihood (C=1, lbfgs, `max_iter`= 1000). The 30,000 rows split 60/20/20 by source row order into 18,000/6,000/6,000, described as row-order rather than temporal since the corpus carries no market-time index.

The linear model is a deployment choice, not an accuracy choice: its coefficients are readable log-odds contributions (Table II) and are exactly the object federated averaging operates on — a weight vector plus intercept with no architectural ambiguity, unlike a tree ensemble, which admits no meaningful coordinate-wise mean. Identical arithmetic in both runtimes is what makes the parity contract testable to 10^{-12}, and the objective yields calibrated probabilities for an interface that presents a probability of need rather than a binary instruction. The accuracy price is quantified in Section IV.

D. Federated Protocol and Aggregation

The parameters of (4) are improved across accounts without any account disclosing its holdings. Fig. 2 shows the round lifecycle: a model is seeded from the reference corpus, distributed to authenticated clients that refuse a contract mismatch, fitted locally in the browser (warm-started from the global parameters under Adam, lr 0.05, batch 32, so the local optimum is a displacement of the global model), and uploaded — twelve coefficients, an intercept, the sample count n_k and the contract versions, nothing else — validated for shape, finiteness and contract match before storage (mismatches rejected as HTTP 409). This scope is stated without ambiguity: this path is **implemented, validated and unit-tested, but disabled by default in the shipped product and not experimentally evaluated in this paper** — no federated accuracy, convergence or robustness result is claimed, and every number in Section IV comes from centralized training. Transmitting updates instead of holdings is data minimisation, not a privacy guarantee; neither differential privacy nor secure aggregation is implemented. Both statements are repeated once in Section V.

Aggregation operates on displacements. Let W_{g} be the active global model; each admitted update is expressed as a delta and clipped in L_2 norm,

$$\Delta_k = W_k - W_{\mathrm{g}}, \qquad \tilde{W}_k = W_{\mathrm{g}} + \Delta_k \min\Big(1,\ \tfrac{L}{\|\Delta_k\|_2}\Big), \quad (5)$$

with L=5.0, then combined by a sample-weighted mean [6],

$$\bar{W} = \sum_k \frac{n_k}{\sum_j n_j} \tilde{W}_k. \qquad (6)$$

Weighting is not cosmetic: two participants contributing 0.4 on 120 samples and 0.8 on 360 samples aggregate to $(120 \cdot 0.4 + 360 \cdot 0.8)/480 = 0.7$ under (6), against 0.6 unweighted. A round executes only under

$$\big|\{k : k \text{ valid}\}\big| \geq 2 \ \wedge \ \text{contract versions current}, \qquad (7)$$

since a single participant would publish its local model as the global one, and mixed contracts would average coefficients defined over different feature or scaler spaces into a vector that is arithmetically well formed and semantically meaningless. Heterogeneous client distributions are a known source of degradation for this family of algorithms [10], [11]; which regime this system falls into is an open question, not a claim made here.

E. Robustness and Parity Contract

The assumed adversary is a participating account, or colluding accounts, attempting to move the global model; the design assumes neither honest participants nor cryptographic identity, so it bounds what an identity is worth rather than preventing Sybil creation: clipping in (5) bounds any single displacement to L=5.0 regardless of submitted magnitude, a cap of $n_k \leq 10{,}000$ bounds the weight one identity can claim in (6) (since n_k is self-reported and unverifiable without seeing data the design is built not to see), and an hourly throttle bounds both submission rate and reconnaissance. Aggregation is administrator-authenticated and lock-protected, activation is push-before-persist and atomic with the ledger, and the append-only ledger makes recovery a forward operation; residual weaknesses (a single static secret, cleartext transport, non-cryptographic identity) are recorded rather than mitigated, and discussed in Section V.

Federated averaging across two runtimes presupposes they compute the same function. Two versioned artefacts enforce this: `feature-spec.json` fixes the feature order and constants of (2); `parity-fixtures.json` pins five reference vectors and their expected standardized forms, logits and probabilities, asserted by both test suites at 10^{-12} (float64) and 10^{-5} (browser float32). The platform refuses to store or aggregate any row whose contract triple differs from the active one (HTTP 409) rather than risk an aggregate silently polluted by incompatible participants — which is also what lets the fixed-constant standardization of (2) stand despite its disclosed scaler-provenance defect, since a scaler refitted per-client would invalidate (6) outright.

TABLE II
THE TWELVE-FEATURE CONTRACT: DEFINITION, DISTRIBUTION OVER THE 30,000-ROW DATASET, UNIVARIATE ASSOCIATION WITH THE LABEL, AND LEARNED COEFFICIENT

#	Feature	Formula	Mean	Std	r with y	Solo AUC	Coef.
Portfolio block							
1	`num_stocks`	N	12.9093	10.0162	-0.0219	0.5235	-0.0352
2	`max_stock_weight`	$w_{\max} = \max_i w_i$	0.2705	0.2287	$+0.0291$	0.5274	$+0.0525$
3	`top3_concentration`	$C_3 = \sum_{\text{top3}} w_i$	0.5253	0.3102	$+0.0216$	0.5273	-0.1686
4	`total_weight_drift`	$D_{\text{ew}} = \sum_i \lvert w_i - \frac{1}{N} \rvert$	0.4092	0.3433	$+0.0331$	0.5321	$+0.0411$
5	`portfolio_return`	$R_p = \sum_i w_i R_i$	0.0570	0.1099	$+0.1979$	0.6051	$+1.0661$
6	`portfolio_volatility`	$V_p = \sum_i w_i (R_i - R_p)^2$	0.0106	0.0209	$+0.0869$	0.5651	-0.0213
7	`sector_concentration`	$S = \max_s \sum_{i \in s} w_i$	0.4386	0.2759	$+0.0239$	0.5170	$+0.0645$
8	`days_since_last_rebalance`	τ (calendar days)	122.6721	67.2327	$+0.0256$	0.5157	-0.0978
Market block							
9	`market_return_30d`	$m_{30} = \text{Cl}_t / \text{Cl}_{t-30} - 1$	0.0144	0.0555	$+0.0441$	0.5417	$+0.0894$
10	`market_volatility_30d`	$\sigma_m = \text{std}_{30}(r)$	0.0090	0.0047	$+0.1086$	0.5975	$+1.0835$
11	`market_drawdown_90d`	$\delta_{90} = (\text{Cl}_t - H_{90}) / H_{90}$	-0.0411	0.0480	-0.0095	0.4972	-0.2602
12	`market_trend`	$\mathbf{1}\{\text{MA}_{20} > \text{MA}_{50}\}$	0.6733	0.4690	$+0.0096$	0.5090	-0.0941

$w_i = q_i P_i / \sum_j q_j P_j$; $R_i = (P_i - P_i^0)/P_i^0$; $\text{Cl}[\cdot]$ are index closes, $r_t = \text{Cl}[t]/\text{Cl}[t-1] - 1$, H_{90} the 90-session peak. Solo AUC: logistic fit on that feature alone. Coefficients: standardized units, intercept 1.6694; the shipped scaler is fitted on a legacy 5,000-row set, not this one (Table III).

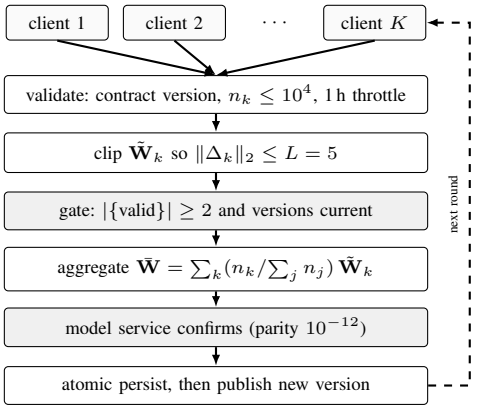


Fig. 2. A federated aggregation round, cf. (5)–(7). Updates are accepted only on the current contract version, L_2-clipped, and persisted only after the model service confirms the aggregate. Implemented and unit-tested; disabled by default and not evaluated here.

IV. IMPLEMENTATION AND RESULT

A. Implementation

The detector of Section III is deployed as the three-tier system of Fig. 1; the evaluation below reproduces the deployed model service's protocol exactly, under the pinned environment (scikit-learn 1.7.2, numpy 2.2.6, pandas 2.3.3, seed 20260804), and every value here was re-executed by a single reproduction script against the committed data. The primary corpus is 30,000 generated rows, split $60/20/20$ in source-row order (test positive rate 0.5027); this is not a temporal split, since the generator emits no timestamp, and the row-shuffled ablation of Table III establishes only that row order introduces no artifact, a weaker property than a genuine temporal holdout. Features use the shipped scaler constants, whose legacy provenance defect is disclosed in Section III and whose cost is quantified below.

B. Performance Evaluation

The primary detector attains test AUC 0.6489 (validation 0.6585, indicating no overfitting) against a chance baseline of 0.5000 and a majority-class accuracy of 0.5027; all other measures (PR-AUC, precision, recall, F_1, MCC, balanced accuracy) are in Table III and each exceeds its baseline. The probability output is usable as a probability, at Brier score 0.2336 (baseline 0.25) and expected calibration error 0.0214 over ten bins. Five-fold cross-validation gives AUC 0.6519 ± 0.0094 and PR-AUC 0.6452 ± 0.0065, and a bootstrap over 1,000 test resamples gives a 95% CI of $[0.6342, 0.6622]$ for test AUC, excluding both the majority-class rate and the random forest's 0.6790 below, so the margin over chance is not an artifact of the test split. At the deployed threshold 0.5, recall is a conservative 0.5212; sweeping the threshold trades this against precision as expected (recall 0.9907 at t=0.3, precision 0.7675 at t=0.7), so a deployment with asymmetric costs should choose its operating point rather than inherit 0.5.

Table III compares the primary model with alternatives on the identical split: regularization strength is inert ($C \in \{0.1, 1, 10\}$ all give test AUC 0.6489), standardization is not (raw features fall to 0.6333), and the best measured model is a random forest at 0.6790, followed by gradient boosting (0.6743) and a calibrated RBF SVM (0.6702) — a deficit of 0.0301 AUC that the logistic detector accepts for the aggregability and calibration argued in Section III.

C. Findings and Comparison with Existing Approaches

The central finding is a matched negative control that varies the label function and nothing else. The same 30,000 feature rows are relabelled by the legacy heuristic scoring rule in place of the backtested certainty-equivalent criterion (3); rows, features, standardization, split, model class and evaluation code are identical across both arms. Fig. 3 shows the result: test AUC 0.9890 under heuristic labels against 0.6489 under backtested labels, a gap of 0.3401 attributable to the label alone. The mechanism is directly observable: elapsed time correlates 0.6074 with the heuristic label, which reads it as an input, and 0.0256 with the backtested label, which contains no elapsed-time term; the two labellings agree on only 0.5221 of rows. The figure 0.9890 is a degeneracy exhibit, not an achievement — it is the score returned when the oracle scored

against is the rule the model was given to imitate — and 0.6489 is the honest difficulty of the task once that oracle is replaced by a forward, net-of-cost outcome. A separate 5,000-row corpus labelled by the same heuristic reaches test AUC 0.9310 and is reported only as corroboration, since its corpus and class balance differ and the shipped scaler is in-sample for it.

Table IV scores the conventional policies against outcome labels on the same test rows. The calendar rule attains AUC 0.5167, the band trigger 0.5146, and — the argumentative payload — the legacy heuristic applied directly as a decision rule attains only 0.5278: the rule that a naively supervised model reproduces at 0.9890 predicts actual net-of-cost benefit barely above chance, so recovering the rule and deciding well are almost unrelated objectives. Paired McNemar tests confirm the ordering in both directions (Table IV): the detector significantly beats every deterministic rule and the heuristic itself ($p < 10^{-16}$ throughout), and is itself significantly beaten by both tree ensembles. Against the alternative learners the comparison runs the other way and is reported as measured: the random forest's 0.0301 AUC advantage is not within noise, but its parameters admit no meaningful coordinate-wise average, whereas the linear model's twelve-weight-plus-intercept vector is precisely what federated averaging operates on. The trade is worth making only because the federated path is pursued; were federation abandoned, a non-federated nonlinear model would be the correct choice on this data.

The fitted coefficients (Table II) are dominated by `market_volatility_30d` (+1.0835) and `portfolio_return` (+1.0661), following directly from the structure of (3), while `top3_concentration` carries a negative coefficient despite a positive univariate correlation — the signature of a concentration family correlated at 0.91–0.92. No economic reading is claimed for the small coefficients.

One caveat applies throughout: 0.6489 is the honest difficulty of the task under outcome labels, not a ceiling — no Bayes-rate estimate was computed, and Table III already shows nonlinear learners exceeding it, so the claim is task hardness, not optimality. This evaluation reports classification metrics only: no path-based backtest was performed, so no Sharpe ratio, Sortino ratio, maximum drawdown, turnover or annualised return is reported anywhere in this paper, and no federated experiment was run — both are future work.

V. Conclusion

This paper treated whether a drifted portfolio should be rebalanced now as a detection problem trained on forward, net-of-cost outcome labels rather than on the heuristic the detector replaces. The detector was not designed first and federated afterwards: because holdings cannot reasonably be centralised and no single participant holds enough decision states, the linear detector and the aggregation design were developed together with the labelling work.

The central result is a matched-control measurement of rule-oracle degeneracy: holding the rows, features, scaler, split,

TABLE III
Model comparison and ablations, on the identical split and feature representation

Model	Val. AUC	Test AUC	Test acc.
Logistic regression ($C = 1$)	0.6585	**0.6489**	0.6022
Random forest (200, depth 8)	0.6750	**0.6790**	0.6290
Gradient boosting (200, depth 3)	0.6729	0.6743	0.6258
SVM-RBF ($C = 1$, calibrated)	0.6773	0.6702	0.6173
Logistic regression, raw features	0.6447	0.6333	0.5942
Majority-class baseline	—	0.5000	0.5027

	Ablation (vs. full 12-feature model, AUC 0.6489)	Test AUC	Δ
A1	without `days_since_last_rebalance`	0.6483	−0.0006
A4	scaler refitted on the training split	0.6490	+0.0001

Regularization strength is inert ($C \in \{0.1, 1, 10\}$ all give test AUC 0.6489); ensembles use `random_state` = 20260804. A1 confirms the backtested label leaves almost no direct time signal, against the 0.6074 correlation the same feature exhibits under heuristic labels (Fig. 3). A4 bounds the disclosed scaler-provenance defect at +0.0001 AUC.

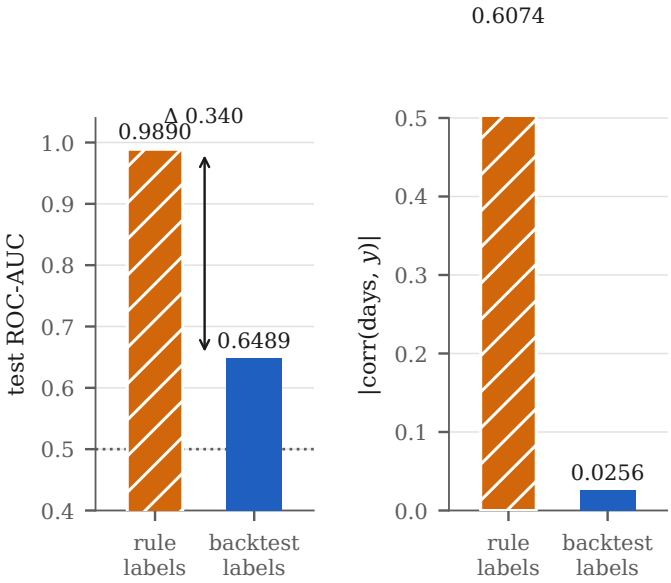


Fig. 3. Rule-oracle degeneracy on one fixed corpus where only the labelling function varies. Left: heuristic labels give test AUC 0.9890 against 0.6489 for backtested labels. Right: the elapsed-time/label correlation rises from 0.0256 to 0.6074 across the same substitution. The high left bar is evidence of a degenerate label, not a capable model.

model class and evaluation code identical and varying only the label function, the same pipeline attains test AUC 0.9890 under heuristic labels and 0.6489 under backtested labels, a gap of 0.3401 attributable to the label rather than the learner (Section IV). Against backtested outcomes the deterministic policies this work replaces carry little information — the heuristic itself predicts realised benefit at only 0.5278 — while the detector is calibrated (ECE 0.0214) and stable under 5-fold cross-validation (0.6519 ± 0.0094).

The federated half is a design contribution of equal standing: updates are L_2-clipped and combined by a sample-weighted mean, gated by a minimum participant count, an upload cap and a per-account throttle, with contract-version admission and atomic activation, and browser/server implementations required to agree to 10^{-12}. This path is implemented and unit-tested, but it is disabled by default and not experimentally

TABLE IV
DETERMINISTIC BASELINES ON THE SAME 6,000 TEST ROWS, AND PAIRED MCNEMAR TESTS AGAINST THE LOGISTIC DETECTOR

	Rule	AUC	Acc.	Prec.	Rec.	MCC
B0	always rebalance	0.5000	0.5027	—	—	0.0000
B1	$\tau \geq 90$ days	0.5167	0.5175	0.5157	0.6598	0.0349
B2	band trigger	0.5146	0.5137	0.5257	0.3319	0.0315
B3	legacy heuristic	0.5278	0.5287	0.5235	0.6953	0.0590
B5	logistic (ours)	**0.6489**	**0.6022**	0.6250	0.5212	0.2079

McNemar exact test vs. tree ensembles (paired hard labels)

Comparison	n_{01}	n_{10}	p
ours vs. random forest	414	575	3.42×10^{-7}
ours vs. gradient boosting	438	580	9.60×10^{-6}

B2 fires when `total_weight_drift` > 0.20 and `max_stock_weight` > 0.35; B3 is the legacy scoring rule of Section III; B1–B3 emit hard decisions, so their AUC equals balanced accuracy. McNemar tests against B0/B1/B2/B3 give $p = 4.48 \times 10^{-24}$, 3.43×10^{-21}, 2.38×10^{-25}, 3.24×10^{-17} respectively (detector significantly better in all four; full n_{01}/n_{10} in the reproduction script). n_{01} counts rows the detector classifies correctly and the comparator does not; n_{10} is the converse — the detector is significantly worse than both tree ensembles.

evaluated in this paper; sharing model updates instead of raw holdings is data minimisation, and no privacy guarantee is claimed.

The limits of the evidence are stated plainly: the cross-section is generated rather than observed, so no external validity is claimed; costs are a flat 35 basis points with no market impact; the $60/20/20$ split is row order with no market-regime holdout; no path-based backtest exists, so no Sharpe, Sortino, drawdown or turnover figure is available; no federated experiment was run; elapsed time is exact here but would be estimated from transaction history for a real user; and the dataset generator resides on an unmerged branch that must ship alongside the reproduction script and pinned dependencies.

Future work follows directly: outcome-derived labels, a market-regime holdout, a multi-factor cross-section and a path-based backtest with turnover and impact modelling would together establish whether the structure learned here survives outside the generator that produced it, and a federated simulation over independent and non-independent partitions, extended to poisoning robustness, should precede any evaluation of differential privacy or secure aggregation. Nothing in this paper constitutes investment advice.

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